

Calculation Supplement — Document 6

Monte Carlo Sensitivity Analysis

TOPSIS weight robustness for the nine-method analysis

343 runs / 9 methods / 4 scenarios

Step-by-step sensitivity analysis

of social navigation in an agricultural digital twin

Purpose. To support the Monte Carlo analysis calculations: $\pm 10\%$ and $\pm 20\%$ weight perturbations, 2000 samples per level, over the nine-method, seven-criterion TOPSIS.

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1 Perturbation model

For nominal weights w , each replicate generates

$$w'_j = w_j(1 + \epsilon_j), \quad \epsilon_j \sim U(-\epsilon_{max}, +\epsilon_{max}), \quad \hat{w}_j = \frac{w'_j}{\sum_k w'_k}. \quad (1)$$

Complete TOPSIS is run with $\hat{w}$, recording each method's rank and CC . The analysis uses $\epsilon_{max} = 0.10$ and 0.20 , with 2000 samples each. The matrix and benefit/cost directions are exactly those of Document 5.

2 Simulation algorithm

For each perturbation amplitude:

1. Fix the normalized matrix and nominal W1 vector.
2. Generate seven independent uniform perturbations.
3. Multiply each weight and renormalize to enforce $\sum_j \hat{w}_j = 1$.
4. Recompute the weighted matrix, ideals, distances, and CC_i .
5. Rank all nine methods and record rank 1, Top-2 membership, mean rank, and CC dispersion.
6. Repeat 2000 times and convert counts into relative frequencies.

For method i , first-rank frequency is

$$f_i^{(1)} = \frac{1}{S} \sum_{s=1}^S \mathbf{1}[r_i^{(s)} = 1], \quad S = 2000. \quad (2)$$

3 Analysis result

The results indicate that under perturbations of primary scheme W1, B4 remains rank 1 in 71–89% of samples. This range covers the two perturbation levels. No complete method-by-level frequency table; this supplement therefore does not invent additional percentages.

Table 1: Monte Carlo result.

Scheme	Configuration	Result
W1	$\pm 10\%$, 2000 samples	B4 reaches rank 1
W1	$\pm 20\%$, 2000 samples	B4 reaches rank 1
Observed combined range:		71–89%

4 Interpretation

B4 frequently, but not absolutely, retains leadership around W1. B3 is a close competitor and leads nominal W3. Leadership changes across schemes, confirming a trade-off surface; sensitivity quantifies local stability around W1 rather than universal superiority.

5 Analytical configuration

The analysis uses nine methods, one 9×7 matrix, three nominal schemes, and 2000 samples per perturbation amplitude. Ranks are computed for each weighting scheme and are not interpreted as a universal ordering.

6 Conclusion

Under W1, B4 is the modal leader in 71–89% of simulations. TOPSIS leadership is frequently robust but remains priority- and perturbation-dependent.

7 Reproducibility control

The pseudorandom seed must be fixed and recorded; the same decision matrix retained; every perturbed vector renormalized; and amplitude, sample count, and nominal scheme reported. Frequencies are computed from integer counts and rounded only for display.